Severity and Case Classification

How to classify a case

Axis	What it describes
Seriousness	Whether the event caused death, was life-threatening, required or prolonged hospitalization, caused disability or a congenital anomaly, or was otherwise medically important. It depends on the outcome or action required.
Severity	The intensity of the reaction. **CTCAE** generally defines Grade 1 as mild, Grade 2 as moderate, Grade 3 as severe, Grade 4 as life-threatening or disabling, and Grade 5 as death. The event-specific criteria determine the grade.
Expectedness	Whether the reaction, including its severity, is described in the applicable reference safety information, such as the SmPC or package insert.
Causality and preventability	How likely the medicine caused the reaction, and whether the event could reasonably have been avoided.

Case 1 Allopurinol and DRESS

A 58-year-old man began allopurinol at 300 mg without gradual dose increases. He developed fever, rash, facial swelling, eosinophilia of 1,400/µL, ALT of 210 U/L, and creatinine of 1.9 mg/dL. He was admitted and treated with steroids, then recovered by week 6.

Axis	Classification and rationale
Seriousness	Serious: hospitalization and a medically important reaction affecting multiple organs.
Severity	Severe; provisionally CTCAE Grade 3. The systemic reaction required hospital-level care; confirm the grade against the event-specific criteria.
Expectedness	Expected: allopurinol product information includes DRESS and hypersensitivity reactions.
Causality	Probable: timing and clinical pattern fit, and he improved after allopurinol was stopped.
Preventability	Potentially preventable: the starting dose was high and was not titrated. Whether HLA-B*58:01 testing was indicated depends on the patient's risk group and applicable guidance.

Case 2 Amoxicillin and A Mild Rash

A 26-year-old woman taking amoxicillin for pharyngitis developed an itchy maculopapular rash on her torso on Day 5. She had no fever, swelling, or breathing difficulty. Amoxicillin was stopped, an antihistamine was given, and the rash cleared within four days. She was treated as an outpatient.

Axis	Classification and rationale
Seriousness	Non-serious: no seriousness criterion was reported.
Severity	Mild to moderate; provisionally CTCAE Grade 1–2. The detailed clinical features determine the grade.
Expectedness	Expected: rash is a known, labelled reaction to amoxicillin.
Causality	Probable: the rash followed treatment and resolved after the medicine was stopped.
Preventability	Not considered preventable from the details provided: there was no known prior penicillin allergy.

Non-serious and expected reactions may still be reported to PvPI, though they are usually a lower priority than serious or unexpected cases.

Case 3 Enalapril and Angioedema

A 62-year-old man had taken enalapril for three weeks when his lips and tongue swelled and he developed stridor. He received intramuscular adrenaline in the emergency department and was monitored overnight in the ICU. Enalapril was stopped, and the swelling resolved within 48 hours.

Axis	Classification and rationale
Seriousness	Serious and life-threatening: airway involvement placed him at immediate risk.
Severity	Severe; provisionally CTCAE Grade 4, given airway compromise and urgent treatment.
Expectedness	Expected: angioedema is a known risk of ACE inhibitors.
Causality	Probable: timing is compatible, symptoms improved after withdrawal, and no other trigger was identified.
Preventability	Not clearly preventable: the reaction can be unpredictable, although some patients have higher risk.

Document the suspected enalapril-related angioedema clearly and avoid ACE inhibitors in future. Any decision about ARB use should be made cautiously by the treating clinician.

Case 4 Methotrexate Error and Fatal Pancytopenia

A 70-year-old woman with rheumatoid arthritis was prescribed methotrexate 7.5 mg weekly. The directions were misread as daily. By Day 10, she had mouth ulcers, pancytopenia, and sepsis. She died in the ICU.

Axis	Classification and rationale
Seriousness	Serious and fatal: the patient died.
Severity	CTCAE Grade 5: the event resulted in death.
Expectedness	Expected: myelosuppression and mucositis are labelled risks, and the product information warns against daily dosing errors.
Causality	Certain or probable: the dosing error and dose-related toxicity provide a strong pharmacological explanation.
Preventability	Preventable: the event followed an error in prescribing, dispensing, or interpreting the weekly schedule.

Report this as both an adverse drug reaction and a medication error. Review system safeguards, including clear weekly-dose labels, pharmacist counselling, and a root-cause review, rather than focusing on individual blame.

Case 5 Hypothetical Zentavir and Acute Liver Failure

A 45-year-old man had taken the newly marketed oral drug Zentavir for four months, with no other known liver-toxic exposures. He developed jaundice, an INR of 3.2, and encephalopathy, and was considered for liver transplantation. He recovered after a transplant. Zentavir and the label details in this teaching example are fictional.

Axis	Classification and rationale
Seriousness	Serious, life-threatening, and medically important: acute liver failure required hospital care and transplant evaluation.
Severity	Severe; provisionally CTCAE Grade 4. Confirm the grade against the event-specific criteria.

Axis	Classification and rationale
Expectedness	Unexpected under the example's assumed label, which describes only mild enzyme elevations and not liver failure. A more severe reaction than described may also be unexpected.
Causality	Probable: the timing is plausible and other causes were reportedly excluded.
Preventability	Not currently considered preventable based on the knowledge provided in this hypothetical case.

This case has the highest regulatory priority in this set because it is serious and unexpected and could indicate a new risk for other patients. In India, CDSCO guidance states that marketing authorization holders must report serious adverse events within 15 calendar days of receiving the information; the applicable requirements depend on the product and regulatory context. A case like this could lead to signal assessment and consideration of a label update or safety communication.

Comparative classification

Case	Drug and reaction	Serious?	Severity	Expected?	Preventability	Priority
1	Allopurinol, DRESS	Yes	Grade 3*	Yes	Potentially	High
2	Amoxicillin, rash	No	Grade 1–2*	Yes	No, based on details	Low
3	Enalapril, angioedema	Yes; life-threatening	Grade 4*	Yes	Not clearly	High
4	Methotrexate, fatal pancytopenia	Yes; fatal	Grade 5	Yes	Yes; error	High
5	Zentavir, liver failure	Yes; life-threatening	Grade 4*	No	No, on current knowledge	Highest; expedited assessment

***Provisional grade:** the exact CTCAE grade depends on the event-specific criteria and complete clinical details.

Ranking by clinical impact

1. Case 4: the patient died, and the medication-use failure could recur without system changes.
2. Case 5: the patient developed life-threatening liver failure, and the unexpected event could reveal a risk affecting other patients.
3. Case 3: the patient had life-threatening airway symptoms, but the risk is known and the episode was treated.
4. Case 1: the patient had a serious systemic reaction and recovered; allopurinol should be avoided in future.
5. Case 2: the rash was self-limiting and treated outside the hospital.

Individual clinical impact and regulatory priority are different. Case 4 had the worst outcome for the patient, while Case 5 ranks highest for regulatory attention because the reaction is serious and unexpected.